classical physics

Toward the end of the 1800s, physicists thought that their understanding of the way that the physical universe worked was nearly complete. Three hundred years earlier, Isaac Newton had described the laws on which the material world operates. In 1864, James Clerk Maxwell revealed the equivalent laws controlling the behavior of light and other electromagnetic phenomena, which apparently completed the story of how the universe of matter and light worked. Yet within a generation, the world of physics was turned upside down by new discoveries of phenomena operating outside the laws of Newton and Maxwell. The "quantum revolution" that these discoveries generated is the one true revolutionary event in the whole history of science. However, to see why it was so revolutionary we need to understand precisely what was overturned by the revolution—the "classical" account of physics according to Newton and Maxwell.

Isaac Newton (1642–1727) laid the foundations of physics with his three Laws of Motion and his Law of Gravity. He also designed an improved astronomical telescope, showed that white light is made up of all the colors of the rainbow, and developed the idea that light is made up of a stream of particles, like tiny cannonballs.

Newton's laws

Among Isaac Newton's discoveries are the three Laws of Motion that describe the way all kinds of objects interact when they collide with each other in the everyday world.

These three laws explain everything from the way atoms and molecules bounce off each other, how the moving parts of a car engine interact, and what's required to send a spacecraft into orbit. Newton's First Law states that any object stays in the same place, or moves in a straight line at a constant speed, unless it is acted on by a force. The truth of this is not obvious in the everyday world because there is always a force—friction—that tends to slow down moving objects. However, objects that are moving in space in free fall, such as planets in orbit around the Sun, flawlessly obey this law because they are moving in a virtual vacuum and there is no friction.

Newton rules
Newton's Laws of Motion are obeyed almost exactly on a nearly frictionless surface. The closest we can come to this on Earth is the perfectly smooth surface of an ice rink. For this reason, the sport of curling is very "Newtonian."

"I do not know what I may appear o the world; but to myself I seem to ave been only like a boy playing on he seashore, and diverting myself in ow and then finding a smoother ebble or a prettier shell than ordinary, vhilst the great ocean of truth lay all ndiscovered before me."

saac Newton

Newton's Second Law says that when a force acts on an object, the acceleration produced is equal to the force applied divided by the mass of the object. This is familiar in everyday life, for example, on a pool table where the harder you hit the white ball, the faster it will move. However, acceleration is not defined solely by an increase in velocity—acceleration can also be defined as a change in direction, or by a combination of changes in both velocity and in direction. So, although the Earth is moving at approximately the same speed, it is also accelerating because the Sun's gravity is

bouncing balls
Collisions between pool balls also obey Newton's laws. In a simple collision between two balls, if one ball is deflected to the right, the other must be deflected to the left.